The maths behind the two notebooks

`nufft_parameter_recovery.ipynb` and `hyperliquid_research.ipynb`

Notes for Mark

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Abstract

Part I derives what the NUFFT notebook computes: the Fourier transform of a point process, why the amplitude and phase of a periodic schedule come out of it, what the null distribution of the power is, why scanning many frequencies needs a corrected threshold, where the "feed comb" comes from, and how clustering (bursts) changes all of it. Along the way it checks the notebook's numbers against closed-form predictions; two of those checks change how some of the notebook's findings should be read (§8). Part II covers the Hyperliquid notebook: the lattice model for block gaps, the autocorrelation tests, the regression and quantile forecasts, the latency model, the hazard model and the event study. Part III lists what changes when the Part I analysis is moved onto the real `market_data_v2` data. No new analysis of real data is in here; the only new numbers are re-computations on the simulation.

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Part I

The NUFFT parameter-recovery notebook

1 The object: a point process with a periodic intensity

The data are event times $0 \le t_1 < t_2 < \cdots < t_N < T$ (trades, quote updates). A *point process* is described by its *intensity* $\lambda(t)$: the expected number of events in a short window $[t, t+\delta)$ is $\lambda(t)\,\delta$. A homogeneous Poisson process has constant λ; "nothing is scheduled".

The notebook plants a schedule as a cosine modulation of the intensity:

$$\lambda(t) \;=\; \lambda_0\big(1 + A\cos(2\pi f_0 t + \varphi)\big), \qquad f_0 = 1/P, \quad 0 \le A \le 1. \tag{1}$$

- λ_0 is the mean rate (`BASE_RATE_HZ`$= 200/\mathrm{s}$),
- P the period (`TRUE_PERIOD_S`$= 1\,\mathrm{s}$), f_0 its frequency,
- A the modulation depth (`TRUE_MOD_A`$= 0.4$): at the busiest phase the rate is $(1+A)\lambda_0$, at the quietest $(1-A)\lambda_0$,
- φ the phase (`TRUE_PHASE`$= 1.10\,\mathrm{rad}$): where in each cycle the activity peaks. The rate is highest when $2\pi f_0 t + \varphi \equiv 0$, i.e. at $t \equiv -\varphi/(2\pi f_0) \pmod P$.

The three numbers to recover are (P, A, φ).

1.1 Simulating it: thinning

`sim_periodic_poisson` uses the Lewis–Shedler thinning method. Since $\lambda(t) \le \lambda_{\max} = \lambda_0(1+A)$:

1. draw a homogeneous Poisson process at rate $\lambda_{\max}$ (a Poisson number of uniform times on $[0, T)$);
2. keep each candidate time t with probability $\lambda(t)/\lambda_{\max} = \dfrac{1 + A\cos(2\pi f_0 t + \varphi)}{1+A}$.

Kept points form a Poisson process with intensity exactly $\lambda(t)$, because independent thinning of a Poisson process with retention probability $p(t)$ gives a Poisson process with intensity $\lambda_{\max} p(t)$.

2 The Fourier transform of a point process

2.1 Definition

For events at t_j with weights w_j (here $w_j = 1$, so we count events), the *non-uniform discrete Fourier transform* (NUDFT) at frequency f is

$$X(f) \;=\; \sum_{j=1}^{N} w_j\, e^{-2\pi \mathrm{i} f t_j} \;=\; \underbrace{\sum_j w_j \cos(2\pi f t_j)}_{C(f)} \;-\; \mathrm{i}\,\underbrace{\sum_j w_j \sin(2\pi f t_j)}_{S(f)}. \tag{2}$$

Each event contributes a unit arrow (a *phasor*) pointing at angle $-2\pi f t_j$. If events are spread evenly around the cycle of period $1/f$ the arrows cancel and $|X(f)|$ is small; if they bunch at one phase the arrows add up.

NUDFT vs NUFFT. Equation (2) *is* the statistic. A NUFFT (e.g. `finufft`) is a fast *algorithm* that evaluates (2) at M frequencies in $O(N \log N + M)$ operations to a chosen accuracy ε, instead of $O(NM)$ for the direct sum. It doesn't change the numbers. The notebook computes the sum directly (`hft_clock.ndft_power`). How a NUFFT does it is described in §3.

2.2 Why the amplitude and phase come out

Take the expectation of (2) at the true frequency. For a Poisson process, $\mathbb{E}\sum_j g(t_j) = \int_0^T g(t)\lambda(t)\,\mathrm{d}t$ (Campbell's theorem). With $g(t) = e^{-2\pi \mathrm{i} f_0 t}$ and writing the cosine as $\frac{1}{2}(e^{\mathrm{i}\theta} + e^{-\mathrm{i}\theta})$:

$$\begin{aligned}\mathbb{E}X(f_0) &= \int_0^T \lambda_0\Big[1 + \tfrac{A}{2}e^{\mathrm{i}(2\pi f_0 t+\varphi)} + \tfrac{A}{2}e^{-\mathrm{i}(2\pi f_0 t+\varphi)}\Big]e^{-2\pi \mathrm{i} f_0 t}\,\mathrm{d}t \\ &= \lambda_0\int_0^T e^{-2\pi \mathrm{i} f_0 t}\,\mathrm{d}t \;+\; \frac{\lambda_0 A}{2}e^{\mathrm{i}\varphi}\int_0^T \mathrm{d}t \;+\; \frac{\lambda_0 A}{2}e^{-\mathrm{i}\varphi}\int_0^T e^{-4\pi \mathrm{i} f_0 t}\,\mathrm{d}t. \end{aligned} \tag{3}$$

The first and third integrals are oscillating terms of size at most $1/(\pi f_0)$, which is negligible when the record holds many cycles ($f_0 T \gg 1$). The middle term grows with T. Since $\mathbb{E}N = \lambda_0 T$:

$$\boxed{\mathbb{E}X(f_0) \approx N\,\frac{A}{2}\,e^{\mathrm{i}\varphi}} \qquad \Longrightarrow \qquad \hat{A} = \frac{2\,|X(f_0)|}{N}, \qquad \hat{\varphi} = \arg X(f_0) = -\operatorname{atan2}(S, C) \bmod 2\pi. \tag{4}$$

This is exactly `recover_at`: `amp_hat = 2*res/total` and `phase_hat = (-arctan2(s, c)) \% 2pi`. The quantity $R = |X(f_0)|/N \in [0, 1]$ is the *Rayleigh resultant* (the length of the average phasor), so $R = A/2$ and $\hat{A} = 2R$.

2.3 The normalised power and its null distribution

`ndft_power` returns

$$z(f) = \frac{|X(f)|^2}{\sum_j w_j^2} \;=\; \frac{|X(f)|^2}{N} \;=\; NR(f)^2 \quad (w_j = 1). \tag{5}$$

Null (no schedule, Poisson). At a frequency with $fT \gg 1$, $C(f)$ and $S(f)$ are sums of N roughly independent terms with mean ≈ 0 and $\mathbb{E}\cos^2 = \mathbb{E}\sin^2 = \frac{1}{2}$. By the central limit theorem they are approximately independent $\mathcal{N}(0, N/2)$. Hence

$$\frac{|X|^2}{N/2} = \frac{C^2 + S^2}{N/2} \sim \chi^2_2 \quad \Longrightarrow \quad z \sim \tfrac{1}{2}\chi^2_2 = \mathrm{Exp}(1), \qquad \Pr(z > x) = e^{-x}. \tag{6}$$

That closed-form tail is why `rayleigh_test` can report `p = exp(-z)` without simulation. This is the *Rayleigh test* of circular uniformity.

With a schedule. From (4), $|X(f_0)|^2 \approx N^2A^2/4$, so

$$z(f_0) \approx \frac{NA^2}{4}. \tag{7}$$

Check against Experiment 1: $N = 360{,}242$, $A = 0.4$ gives $z \approx 14{,}410$; the notebook reports 14,350.

2.4 How precise are $\hat{A}$ and $\hat{\varphi}$?

Around its mean, $X(f_0)$ has noise of variance $N/2$ in each of the two directions. Projecting onto the direction of the mean and onto the direction at right angles to it:

$$\mathrm{se}(\hat{A}) \approx \frac{2}{N}\sqrt{\frac{N}{2}} = \sqrt{\frac{2}{N}}, \qquad \mathrm{se}(\hat{\varphi}) \approx \frac{\sqrt{N/2}}{|X|} = \frac{1}{R\sqrt{2N}} = \frac{\sqrt{2}}{A\sqrt{N}}. \tag{8}$$

For Experiment 1, $\mathrm{se}(\hat{A}) \approx 0.0024$ and $\mathrm{se}(\hat{\varphi}) \approx 0.006\,\mathrm{rad}$. The observed errors (0.0003–0.0008 in A) fit that. The NUDFT *phase* error of 0.118 rad is far larger, and comes from the next subsection, not from noise.

2.5 Frequency resolution, leakage, and the phase penalty for a slightly wrong frequency

Evaluate $\mathbb{E}X$ at a frequency $f = f_0 + \delta$ close to, but not equal to, the truth. Repeating the calculation above, the middle term becomes

$$\mathbb{E}X(f_0+\delta) \approx \frac{\lambda_0 A}{2} e^{\mathrm{i}\varphi} \int_0^T e^{-2\pi \mathrm{i}\delta t}\,\mathrm{d}t \;=\; N\frac{A}{2}\,e^{\mathrm{i}\varphi}\;\underbrace{e^{-\mathrm{i}\pi\delta T}}_{\text{phase drift}}\;\underbrace{\operatorname{sinc}(\delta T)}_{\text{leakage}}, \qquad \operatorname{sinc}(x) = \frac{\sin \pi x}{\pi x}. \tag{9}$$

Three consequences:

1. **Line width (resolution).** The peak has a sinc shape in frequency, falling to zero at $\delta = \pm 1/T$. So $1/T$ is the frequency resolution: $1/1800 \approx 5.6 \times 10^{-4}\,\mathrm{Hz}$ in the notebook. Two periods closer than this in frequency can't be told apart.
2. **Sidelobes (leakage).** Away from the peak $|\operatorname{sinc}(\delta T)| \approx 1/(\pi|\delta|T)$. A strong line puts power into frequencies many resolution cells away. The power sidelobes decay only like $1/(\pi\delta T)^2$.
3. **Phase drift.** If you estimate the phase at a slightly wrong frequency, the phase is biased by $-\pi\delta T$. In Experiment 1 the NUDFT frequency estimate was off by 0.0021%, i.e. $\delta = 2.1 \times 10^{-5}\,\mathrm{Hz}$, which predicts a phase bias of $\pi \cdot 2.1 \times 10^{-5} \cdot 1800 = 0.119\,\mathrm{rad}$. The notebook shows 0.118. The FFT's grid contains exactly 1 Hz ($= 1800/T$), so it had $\delta = 0$ and no phase error.

Rule. Scan frequencies at a spacing of at most $\sim 1/(4T)$ (or refine around every candidate), and estimate the phase at a well-pinned frequency. If you name the period in advance, evaluate exactly there.

3 Uniform grids, the FFT, and what a NUFFT does internally

3.1 Binning onto a grid

`grid_counts` histograms the events into $n = T/\Delta$ bins of width $\Delta = 5\,\mathrm{ms}$ with centres τ_k, giving counts c_k. The ordinary DFT of the counts at the FFT frequencies $f_m = m/(n\Delta) = m/T$ is

$$\tilde{X}(f_m) = \sum_{k=0}^{n-1} c_k\, e^{-2\pi \mathrm{i} f_m \tau_k} = \sum_j e^{-2\pi \mathrm{i} f_m \tau_{k(j)}}, \tag{10}$$

so it is the NUDFT (2) with every event moved to the centre of its bin. Write $t_j = \tau_{k(j)} + u_j$ with offsets $u_j \in [-\Delta/2, \Delta/2)$. If the offsets are spread uniformly, which is true for any smooth process,

$$\mathbb{E}[e^{+2\pi \mathrm{i} f u}] = \frac{1}{\Delta}\int_{-\Delta/2}^{\Delta/2} e^{2\pi \mathrm{i} f u}\,\mathrm{d}u = \operatorname{sinc}(f\Delta) \quad\Longrightarrow\quad \tilde{X}(f) \approx \operatorname{sinc}(f\Delta)\, X(f). \tag{11}$$

Binning *attenuates* a line at frequency f by $\operatorname{sinc}(f\Delta)$. At $f = 1\,\mathrm{Hz}$ and $\Delta = 5\,\mathrm{ms}$ that is 0.99996, so it doesn't matter; at f near $1/\Delta$ it does. Three more facts about the grid:

- **Nyquist.** A grid of spacing Δ can represent frequencies only up to $1/(2\Delta) = 100\,\mathrm{Hz}$. Power above that *aliases* back to lower frequencies.

- **FFT frequencies.** `np.fft.rfft` returns $\tilde{X}$ only at $f_m = m/T$, exactly one resolution cell apart. That's the minimum sampling that keeps all the information, but a line between two f_m is attenuated by up to $\mathrm{sinc}(1/2) = 0.64$ in amplitude (0.41 in power). Zero-padding the counts samples in between.
- **Units.** `fft_spectrum` returns $|\tilde{X}|^2$ unnormalised, and `ndft_power` returns $z = |X|^2/N$. Their heights aren't comparable; only the locations and the recovered (A, φ) are.

3.2 Gridding + deconvolution (`scan_band`) and the real NUFFT

`hft_clock.scan_band` bins at $\Delta = P_{\min}/20$, runs an FFT, and divides by $\mathrm{sinc}(f\Delta)$ to undo (11):

$$z(f_m) = \frac{\left|\tilde{X}(f_m)/\,\mathrm{sinc}(f_m\Delta)\right|^2}{N}. \tag{12}$$

This is the crudest member of the NUFFT family. A proper type-1 NUFFT:

1. *spreads* each event onto an oversampled grid (typically 2×) with a smooth, compact kernel ϕ (Gaussian, Kaiser–Bessel or "exponential of semicircle") instead of a box;
2. FFTs the grid;
3. divides by the kernel's Fourier transform $\hat{\phi}(f)$.

The smooth kernel makes the aliasing error decay exponentially with kernel width, so accuracy is controlled ($\varepsilon \sim 10^{-6}$ to 10^{-12}), whereas a box kernel only gives the sinc correction on average.

In the code `verify_grid` recomputes the exact sum at a few frequencies to check the gridded scan. Ratios near 1 mean the gridding error is negligible.

4 Scanning many frequencies: the multiplicity threshold

A scan over the band $[f_{\mathrm{lo}}, f_{\mathrm{hi}}]$ looks at about

$$M \approx T\,(f_{\mathrm{hi}} - f_{\mathrm{lo}}) \tag{13}$$

independent frequencies, one per resolution cell. Under the null each $z \sim \mathrm{Exp}(1)$, so the largest of M independent ones satisfies

$$\Pr\Big(\max_{m \le M} z_m > x\Big) = 1 - (1 - e^{-x})^M \approx M e^{-x}. \tag{14}$$

Setting this equal to the family-wise level α gives the threshold

$$z^* = \ln \frac{M}{\alpha}. \tag{15}$$

In Experiment 4 the band is 0.2–20 Hz over $T = 1800$ s: $M = 35{,}640$, $z^* = \ln(3.56 \times 10^6) = 15.1$, which is `z_thresh_01`. On real data spanning a day this grows only logarithmically, but the *typical* null maximum is about $\ln M$, so "look at that spike" is worthless without the line drawn.

Named candidates (`candidate_test`) avoid this: with $M \approx 10$ pre-named periods, Bonferroni multiplies each p by 10, and that's all.

5 Bursts: why the Exp(1) null breaks

Real order flow is clustered: one event triggers more. `add_bursts` uses a Neyman–Scott cluster process. Each *parent* spawns $C \sim \mathrm{Poisson}(m)$ children (m=`BURST_CHILDREN`= 2.5) at delays $D \sim \mathrm{Exp}$ with mean d (`BURST_DECAY_S`= 20 ms).

5.1 Variance inflation of the spectrum

Parents are Poisson, so $X(f)$ is a sum of independent *cluster* phasors $Y = e^{-2\pi \mathrm{i} f t_p}(1+\sum_{i=1}^{C} e^{-2\pi \mathrm{i} f D_i})$. Let

$$\psi(f) = \mathbb{E}\, e^{-2\pi \mathrm{i} f D} = \frac{1}{1 + 2\pi \mathrm{i} f d} \tag{16}$$

be the characteristic function of the delay. Using $\mathbb{E}C = m$ and $\mathbb{E}C(C-1) = m^2$ (Poisson),

$$\mathbb{E}|1 + \textstyle\sum_i e^{-2\pi \mathrm{i} f D_i}|^2 = 1 + m + 2m\,\mathrm{Re}\,\psi(f) + m^2|\psi(f)|^2. \tag{17}$$

Dividing by the mean cluster size $1+m$ (because N counts children too) gives the factor by which the null variance of X exceeds N:

$$\boxed{\kappa(f) = \frac{1 + m + 2m\,\mathrm{Re}\,\psi(f) + m^2|\psi(f)|^2}{1+m}, \qquad z(f) \;\sim\; \kappa(f)\cdot \mathrm{Exp}(1) \text{ under the null.}} \tag{18}$$

With the notebook's knobs:

f (Hz)	$\to 0$	1	4	20
$\kappa(f)$	4.21	4.16	3.57	1.44

So at low frequencies the null spectrum is about four times taller than Exp(1) says. The analytic p-value and the threshold (15) are *anticonservative* there by roughly that factor. That's why the null in Experiment 4 throws its biggest false peak at the *long-period* end of the band ($P = 4.11\,\mathrm{s}$, $z = 33.8$). At that end of the band the typical null maximum is of order $\kappa \ln M \approx 4.2 \times 10.5 \approx 44$, so 33.8 is an unremarkable draw.

5.2 The cure: block permutation

`shuffle_null` keeps the empirical gaps between events but permutes *blocks* of consecutive gaps. Within a block the clustering survives, so κ is preserved; across blocks the alignment to the wall clock is destroyed, so the schedule is removed. Its weakness: clusters longer than the block get cut, and blocks much longer than a cluster make the null too easy to match. Hence the block-length sweep in `run_analysis`.

5.3 Bursts also bias the phase and amplitude (a deterministic effect)

Children inherit their parent's phase but arrive D later. At the signal frequency the expected phasor gets multiplied by

$$\gamma(f) = \frac{1 + m\,\psi(f)}{1+m}, \qquad \hat{A} \to A\,|\gamma(f_0)|, \qquad \hat{\varphi} \to \varphi + \arg\gamma(f_0). \tag{19}$$

Here $\arg\gamma < 0$, so $\hat{\varphi}$ comes out lower by $|\arg\gamma|$. That means the activity peaks $|\arg\gamma|/(2\pi f_0)$ seconds *later* in the cycle, and the amplitude is shrunk. That's not estimation error; the simulated flow really does peak later than the parents' schedule. Table 1 compares this prediction with Experiment 3's "phase error at the named candidate": The rising phase error at short periods is fully explained by $\arg\gamma$. The NUDFT and FFT agree with each other (and with the prediction) because they're estimating the same, shifted, quantity. Likewise, Experiment 2's $\hat{A} = 0.3855$ and $\hat{\varphi} = 1.021$ match the prediction $A\,\frac{N_{\mathrm{HL}}}{N}|\gamma(1)| = 0.385$ and $\varphi + \arg\gamma(1) = 1.011$ to within one standard error. (The Binance leg dilutes the amplitude, see §8.)

Table 1: Experiment 3 phase errors vs the closed-form burst bias $|\arg\gamma(1/P)|$ from (19).

P (s)	f_0 (Hz)	predicted $\lvert\arg\gamma\rvert$ (rad)	notebook `nu_pherr` (rad)	$\lvert\gamma\rvert$ (amplitude factor)
0.08	12.50	0.582	0.586	0.589
0.15	6.67	0.462	0.484	0.788
0.25	4.00	0.323	0.330	0.903
0.50	2.00	0.175	0.171	0.972
1.00	1.00	0.089	0.095	0.993
2.00	0.50	0.045	0.049	0.998
3.00	0.33	0.030	0.022	0.999

6 The feed comb

6.1 Why a lattice shows up as a comb

Suppose events sit on a lattice $t = n\Delta_B + \epsilon_n$ (Binance's 100 ms cadence, or Hyperliquid's $\sim$67 ms blocks), with small independent jitter ϵ_n. Ignoring jitter, the transform of a full lattice is

$$\sum_n e^{-2\pi i f n\Delta_B} \;\propto\; \sum_{h\in\mathbb{Z}} \delta\Big(f - \frac{h}{\Delta_B}\Big) \qquad \text{(Poisson summation)}, \tag{20}$$

a *comb* of lines at $f = h/\Delta_B$, i.e. periods $\Delta_B, \Delta_B/2, \Delta_B/3, \ldots$ With N_B lattice events (all present, or thinned at random) and jitter characteristic function ϕ_ϵ:

$$\mathbb{E}X_B\Big(\tfrac{h}{\Delta_B}\Big) \approx N_B\,\phi_\epsilon\Big(\tfrac{h}{\Delta_B}\Big), \qquad z_{\text{comb}}(h) \approx \frac{N_B^2\,|\phi_\epsilon(h/\Delta_B)|^2}{N}. \tag{21}$$

The comb's power scales like N_B^2/N, the square of the number of lattice events, and it carries no information about traders. Jitter only damps the high harmonics:

- exponential jitter of scale s: $|\phi_\epsilon(f)| = 1/\sqrt{1+(2\pi f s)^2}$; for $s = 0.3$ ms at 20 Hz that's 0.9993, so no damping;
- Gaussian jitter of s.d. σ: $|\phi_\epsilon(f)|^2 = e^{-4\pi^2\sigma^2 f^2}$.

6.2 Comb exclusion

`comb_keep_mask` masks every period within $\pm 3\%$ of Δ_B/h for $h = 1, \ldots, 25$. That's crude but honest. Anything at, or harmonically related to, the feed's own clock is presumed to be the feed.

7 Local contrast, phase stability, and why scan-and-select fools them

Local contrast (`local_contrast`) is the peak power over the median power of neighbours within $\pm 25\%$ in frequency. It is immune to a broad slope in the spectrum (such as $\kappa(f)$ above).

Phase stability (`stability_check`) splits the record into $W = 6$ windows, estimates R_w and phase θ_w in each, and reports

$$\text{phase agreement} = \frac{\big|\sum_w R_w e^{i\theta_w}\big|}{\sum_w R_w} \in [0,1]. \tag{22}$$

Why selection fools both. The full-record transform is the *sum* of the window transforms, $X = \sum_w X_w$. Picking the frequency that maximises $|X|$ out of 35,640 therefore picks a frequency

where the window phasors *happen* to point the same way, so phase agreement comes out high *by construction*. Likewise the selected peak is, by construction, taller than its neighbours. The tests are still useful for a period named in advance; they are not independent confirmations of a period you selected. What survives is *effect size*. The planted schedule had $z \approx 21{,}700$ versus 34 for the selected null peak.

8 Checks on the notebook's numbers

I re-ran the simulation (same seed and draw order) to check the closed-form predictions above. Two results change the interpretation of the notebook's findings.

1. The "feed-comb trap" in Experiment 2 is mainly a scan-grid artefact. `nudft_spectrum` uses `N_SCAN_FREQ = 4000` frequencies over 0.2–20 Hz, a step of 0.00495 Hz, which is 8.9 resolution cells. The true 1 Hz line is only one cell wide, so the scan steps over it:

line	nearest scan frequency (offset in cells)	z at scan point	exact z at the line
schedule, 1 Hz	1.00210 (+3.78)	54.6	**21,727**
comb, 10 Hz	9.99850 (−2.70)	2.2	530
comb, 20 Hz ($=f_{\text{hi}}$)	20.00000 (0.00)	568	568

The comb "wins" only because 20 Hz is the band *endpoint*, which `linspace` hits exactly. On a fine enough grid the schedule is about 40× taller than the comb. The FFT didn't fall into this because its frequencies m/T include 1 Hz exactly. Experiment 3 uses 1500 frequencies per band, and its offsets grow as $1/P$ (9.0 cells at $P = 80$ ms, 2.9 at 250 ms). That is the likely cause of the NUDFT period misses at those two periods. I haven't re-run those two with a finer grid, so I'm stating it as likely rather than shown. *Fix:* step $\leq 1/(4T)$, or refine around the top few candidates, and keep band edges off comb harmonics.

2. The Binance leg isn't 55% of events; it's 3.1%, and it carries no schedule. `sim_hl_binance` sets the fire probability to `n_bin_budget/grid.size`$\times(1+A\cos\cdot) = 198{,}000/18{,}000\times(\dots) = 11\times(\dots)$, which is then clipped to 1. Every one of the 18,000 grid points fires, regardless of the schedule. So the merged stream is 566,755 bursty HL events plus a *full* 100 ms lattice of 18,000 unmodulated events (`BIN_SHARE` isn't realised). Prediction (21): $z_{\text{comb}} \approx 18{,}000^2/584{,}755 = 554$, matching the 530–568 observed. The lattice also dilutes $\hat{A}$ by $566{,}755/584{,}755 = 0.969$.

Neither issue affects the core methodological points: name candidates in advance, mask the comb, judge by effect size, and use a burst-preserving null. They do change two headline sentences. "The NUDFT is captured by the comb" should read "a coarse frequency grid can step over a narrow line". "The comb drags the NUDFT peak near the feed scale" needs re-testing with a fine grid.

Part II

The Hyperliquid block-timing notebook

9 Setup and notation

- Latent HL blocks are committed at times $b_1 < b_2 < \dots$ with *block intervals* $\beta_i = b_i - b_{i-1}$. The feed has no block height, so b_i is never observed directly.
- A block is *visible* if at least one of the 6 instruments printed an l1 (bbo) update or trade with that `ts_exch` (ms). Visible blocks $t_1 < t_2 < \dots$ are a *thinned* subset of the b_i.
- The *observed gap* $g_n = t_n - t_{n-1}$ is the sum of $k_n \geq 1$ consecutive block intervals ($k_n - 1$

invisible blocks in between):

$$g_n = \sum_{i=1}^{k_n} \beta_{(n,i)}. \tag{23}$$

- Gaps that span a websocket reconnect are dropped; trade replays on resubscribe are removed.
- Train/test split: the first 60% of the record by time is train (`TRAIN_FRAC`). Everything is fit on train and scored on test.

10 Q1: what distribution do gaps follow?

10.1 Discretised models

Timestamps are integer ms, so every model is turned into a probability mass function (pmf) on $g \in \{1, \dots, G_{\max}\}$ ($G_{\max} = 600$) by

$$p(g) \propto F(g + \tfrac{1}{2}) - F(g - \tfrac{1}{2}) \tag{24}$$

for a continuous cdf F (`_disc`). The models:

- **M0 exponential**, $F(x) = 1 - e^{-x/s}$: blocks arrive as a Poisson process, the "no clock" null.
- **M1 shifted lognormal**: $\log(g - c) \sim \mathcal{N}(\mu, \sigma^2)$, a smooth one-humped duration.
- **M2a rigid lattice**: one block $\beta \sim \mathcal{N}(\mu, \sigma^2)$; gap = sum of k blocks, with free weights w_k.
- **M2b lattice with slow blocks**: one block is a two-part mixture

$$b(\beta) = \pi\, \mathcal{N}(\beta; \mu, \sigma^2) + (1 - \pi)\, \mathrm{Gamma}(\beta - c;\, a,\, \theta), \tag{25}$$

  i.e. a sharp clock component plus a spread-out component for blocks that run long or short.

10.2 From one block to a gap: convolution

If the k block intervals in (23) are independent with pmf b, the pmf of their sum is the k-fold convolution b^{*k} ($b^{*1} = b$, $b^{*k} = b^{*(k-1)} * b$, computed with `np.convolve`). Mixing over k:

$$p(g) = \sum_{k=1}^{K} w_k\, b^{*k}(g), \qquad w_k = \frac{e^{\eta_k}}{\sum_\ell e^{\eta_\ell}} \quad (K = 8). \tag{26}$$

The mixture weights use a softmax and π a logistic (`expit`); σ, a and θ are fitted on the log scale. That makes every parameter unconstrained, so L-BFGS can be used.

> **Assumption:** k is independent of the block lengths. It isn't exactly: a long block has more time to collect a visible change. So the w_k are descriptive, not structural.

10.3 Maximum likelihood on a histogram

With h_g the number of train gaps equal to g, the log-likelihood of i.i.d. gaps is multinomial:

$$\ell(\theta) = \sum_g h_g \log p_\theta(g), \qquad \hat{\theta} = \arg\max_\theta \ell(\theta). \tag{27}$$

Fitting on the histogram is exact (not an approximation) and makes the cost independent of n. The notebook reports the test negative log-likelihood per observation, $-\ell_{\text{test}}(\hat{\theta})/n_{\text{test}}$, so models are compared out of sample.

10.4 Goodness of fit: Pearson χ^2

On test, the expected count in bin g is $E_g = n_{\text{test}}\, p_{\hat\theta}(g)$. Bins are merged left to right until each merged bin has $E \geq 5$, and then

$$\chi^2 = \sum_{\text{bins}} \frac{(O-E)^2}{E}, \qquad \text{dof} = \#\text{bins} - 1 - \#\text{params}. \tag{28}$$

With $n \approx 80{,}000$, any small misfit gives $p \approx 0$. Read χ^2/dof and the maximum cdf gap as *sizes* of misfit.

10.5 Derived quantities

- **Posterior number of blocks in a gap.** Bayes on (26): $\Pr(k = 1 \mid g) = w_1 b(g)/p(g)$. The single-block cut `SINGLE_MAX` is the largest g above the mode at which this is still ≥ 0.9 (99 ms).
- **Mean block** $\bar\beta = \sum_\beta \beta\, b(\beta)$ (68.1 ms). Blocks per second $= 1000/\bar\beta$.
- **Share of blocks visible** $\approx \#\text{visible}/((t_{\max} - t_{\min})/\bar\beta)$.
- **Per instrument**, $k \approx \text{round}(g/\bar\beta)$: how many blocks apart that instrument's updates are.

11 Q2: are gaps independent?

11.1 ACF with gaps, and Ljung–Box

For a series x_n with mean $\bar x$, the lag-k autocorrelation is

$$\hat\rho(k) = \frac{\frac{1}{|\mathcal{P}_k|}\sum_{(n,n+k)\in\mathcal{P}_k}(x_n - \bar x)(x_{n+k} - \bar x)}{\frac{1}{n}\sum_n (x_n - \bar x)^2}, \tag{29}$$

where $\mathcal{P}_k$ is the set of pairs where both values exist (`acf_nan`). That's how "single-block gap" works: non-single gaps are set to NaN, so they drop out pair by pair without shifting the series. Under H_0 (white noise), $\hat\rho(k) \approx \mathcal{N}(0, 1/n)$, giving the $\pm 1.96/\sqrt{n}$ band. Ljung–Box combines lags $1..L$:

$$Q(L) = n(n+2)\sum_{k=1}^{L} \frac{\hat\rho(k)^2}{n-k} \;\sim\; \chi^2_L \quad \text{under } H_0. \tag{30}$$

At $n \approx 200{,}000$ the band is ± 0.004, so tiny ρ is "significant". Read magnitudes.

11.2 Three series, because raw gaps mix two processes

- **raw gap** g_n: clock and visibility mixed;
- $k_n = \text{round}(g_n/\bar\beta)$: visibility alone (bursty periods have many visible blocks in a row, giving positive ρ);
- **single-block gap**: g_n if $g_n \leq 99$, else NaN; this is the clock alone.

11.3 What would ms rounding alone do?

Write the recorded time $t_n = t_n^* + e_n$ with rounding error $e_n \sim U(-\frac{1}{2}, \frac{1}{2})$ ms, independent across n, $\text{Var}\, e = 1/12$. Then $g_n = g_n^* + e_n - e_{n-1}$, so if the true gaps are uncorrelated:

$$\text{Cov}(g_n, g_{n+1}) = \text{Cov}(e_n - e_{n-1},\, e_{n+1} - e_n) = -\,\text{Var}(e) \quad\Longrightarrow\quad \rho(1) \approx -\frac{1/12}{\text{Var}(g)}. \tag{31}$$

With the notebook's single-block $\mathrm{Var}(g)$ (tens of ms^2) this is ≈ -0.001. The observed -0.20 is a property of the clock, not of rounding.

11.4 Interpreting $\rho(1) = -0.2$: free-running vs phase-locked clock

Two simple models bracket it:

- **Free-running**: each block takes $\mu + \xi_n$ after the last, ξ i.i.d. Gaps are i.i.d., so $\rho(1) = 0$.
- **Phase-locked with jitter**: block n is scheduled at $n\mu$ but lands at $n\mu + e_n$ with e_n i.i.d. Then $g_n = \mu + e_n - e_{n-1}$ and, as above, $\rho(1) = -\operatorname{Var}(e)/(2\operatorname{Var}(e)) = -\frac{1}{2}$, with $\rho(k) = 0$ for $k \geq 2$.

A mixture of the two (some catch-up toward a schedule) gives $-\frac{1}{2} < \rho(1) < 0$. The observed -0.2 and the table "prev $\leq 55 \Rightarrow$ next median 73" are consistent with partial catch-up. This is a description, not a test of mechanism.

12 Q3: does activity change timing? Mann–Whitney U

Split blocks into "high" (activity above the train 80th percentile) and "low". With samples Y^{hi} and Y^{lo} of the next gap,

$$U = \sum_i \sum_j \Big[\mathbb{1}(Y_i^{\mathrm{hi}} > Y_j^{\mathrm{lo}}) + \tfrac{1}{2}\mathbb{1}(Y_i^{\mathrm{hi}} = Y_j^{\mathrm{lo}}) \Big], \qquad \mathrm{AUC} = \frac{U}{n_{\mathrm{hi}} n_{\mathrm{lo}}} = \Pr(Y^{\mathrm{hi}} > Y^{\mathrm{lo}}). \tag{32}$$

H_0 (same distribution) gives AUC $= \frac{1}{2}$. The notebook also reports the median difference with a 300-draw bootstrap CI. Values are integer ms, so ties are common; SciPy's U test corrects for ties. The single-block-only target conditions on the next gap being short, so it describes the clock rather than giving a forecast.

13 Q4: do lags predict the next gap? OLS + HAC Wald

Regress $y_n = \log g_{n+1}$ on $x_n = (1, \log g_n, \ldots, \log g_{n-9})$:

$$y_n = x_n^\top \beta + u_n, \qquad \hat{\beta} = (X^\top X)^{-1} X^\top y. \tag{33}$$

HAC covariance. The errors u_n are autocorrelated and heteroskedastic, so the usual $\hat{\sigma}^2 (X^\top X)^{-1}$ is wrong. Newey–West uses

$$\hat{V} = (X^\top X)^{-1} \hat{S} (X^\top X)^{-1}, \quad \hat{S} = \sum_{|\ell| \leq L} \Big(1 - \tfrac{|\ell|}{L+1}\Big) \sum_n \hat{u}_n \hat{u}_{n-\ell}\, x_n x_{n-\ell}^\top, \quad L = 20. \tag{34}$$

Wald test of $H_0 : R\beta = 0$, where R picks the tested coefficients (q of them):

$$W = (R\hat{\beta})^\top (R \hat{V} R^\top)^{-1} (R\hat{\beta}) \mathrel{\dot{\sim}} \chi^2_q, \qquad F = W/q. \tag{35}$$

Out-of-sample R^2 compares the test-set squared error with that of the *train* mean: $R^2_{\mathrm{OOS}} = 1 - \sum_{\mathrm{test}} (y - \hat{y})^2 / \sum_{\mathrm{test}} (y - \bar{y}_{\mathrm{train}})^2$. It can be negative.

14 Features and the truncation test

Every feature at block n must be a function of data with timestamp $\leq t_n$ (*point-in-time*). The truncation test checks this mechanically. For a cut c:

$$F_n(\mathcal{D}) = F_n(\mathcal{D}_{\leq c}) \quad \text{for every } n \text{ with } t_n \leq c, \tag{36}$$

where $\mathcal{D}_{\le c}$ deletes every event after c. Any feature that peeks forward fails this at some cut. Window counts use half-open windows $(a, b]$ via `searchsorted`, so an event exactly at t_n counts as known at t_n.

15 Q5: forecasting within an interval, with latency

15.1 Latency model

Let $L_m \ge 0$ be the delay before block m is known to us. Block n is usable only once it *and every earlier block* is known, so the decision time is

$$K_n = \max_{m \le n} (t_m + L_m) \qquad \text{(prefix max, } \texttt{np.maximum.accumulate}\text{)}. \tag{37}$$

Staleness is $K_n - t_n$. The **target** is the residual wait, in exchange time, until the first visible block after we know about block n:

$$y_n = t_{j(n)} - K_n, \qquad j(n) = \min\{j : t_j > K_n\}. \tag{38}$$

With $L \approx 0$ this is just the next gap. Binance features at K_n use only Binance events whose own known time (their t plus latency) is $\le K_n$.

Scenarios: (i) $L \sim \mathcal{N}(0.5,\, 0.05^2)$ ms, clipped at 0; (ii) measured $L_m = \max(\texttt{ts_recv}) - t_m$ over the block's messages.

15.2 Why 0.5 ms end-to-end is ruled out by the data

Both venues are stamped by the same VM clock. For a message, $\texttt{recv} - \texttt{exch} = L + \delta_{\text{venue}}$, where δ is the VM clock's offset from that venue's clock. Since $L \ge 0$, $\min(\texttt{recv} - \texttt{exch}) \ge \delta$. Observed minima: HL $\approx$ 197 ms, Binance $\approx$ 51 ms. If HL's L were 0.5 ms, the VM would be $\delta \approx 196$ ms ahead of HL. Unless HL's and Binance's clocks disagree by about 150 ms, Binance latency would then be negative. So L_{HL} is about 150–280 ms, unless the two exchange clocks are far apart.

15.3 Quantile forecasts and the pinball loss

For a level $\tau \in (0, 1)$, the *pinball* (check) loss of a forecast q for outcome y is

$$\rho_\tau(y - q) = \max(\tau(y - q),\, (\tau - 1)(y - q)). \tag{39}$$

Its expected value is minimised by the τ-quantile of y: the derivative in q is $\Pr(y < q) - \tau$, which is zero at $q = F^{-1}(\tau)$. So fitting a model by minimising pinball loss gives a conditional quantile. The notebook fits gradient-boosted trees (`HistGradientBoostingRegressor`, `loss="quantile"`) for $\tau \in \{0.1, 0.5, 0.9\}$. It sorts the three predictions to stop the quantiles crossing, and $[q_{0.1}, q_{0.9}]$ is the 80% interval.

Metrics. With $\bar{L}_F$ the mean pinball loss over the three τ for feature set F:

$$\text{skill}_F = 1 - \frac{\bar{L}_F}{\bar{L}_{F_0}}, \qquad \text{gain}_{F_j} = \frac{\bar{L}_{F_{j-1}} - \bar{L}_{F_j}}{\bar{L}_{F_0}}, \tag{40}$$

where F_0 predicts the unconditional train quantiles. Also reported: coverage $= \Pr(q_{0.1} \le y \le q_{0.9})$ (target 0.8), median width, and MAE of $q_{0.5}$.

15.4 Moving-block bootstrap CI

Per-observation loss differences d_n are autocorrelated, so the i.i.d. bootstrap would understate uncertainty. The moving-block bootstrap resamples contiguous blocks of length $\ell = 500$ (random starts) until n values are drawn, recomputes $\bar{d}$, repeats 400 times, and takes the 2.5/97.5 percentiles. It is valid if dependence dies out well within ℓ observations.

16 Binance inside HL's gaps

16.1 Discrete-time hazard

Cut each gap into $\Delta = 10\,\text{ms}$ steps $e = 0, 10, 20, \ldots$ (up to 600 ms). The hazard is

$$h(e \mid x) = \Pr\big(G \in (e, e+\Delta] \mid G > e,\ x\big). \tag{41}$$

A gap that ends in step J contributes $\prod_{e<J}(1-h(e)) \cdot h(J)$ to the likelihood. That is exactly a product of independent Bernoulli terms, one per (gap, step) row with outcome 1 only in the final step (gaps longer than 600 ms are censored: all their rows are 0). So any binary classifier fitted on those rows estimates h, and the row-wise log-loss is the negative log-likelihood. Model A uses HL information at block n plus elapsed time e; model B adds Binance activity since block n and in the last 20 ms. The gain is $\Delta\text{LL} = \text{LL}_A - \text{LL}_B$, bootstrapped over gaps.

16.2 Clock test

Spearman's ρ_S is the Pearson correlation of ranks, so it tests a monotone association. It is computed between Binance updates in the first 40 ms after block n and the next gap, among single-block next gaps only.

16.3 Lead profile and within-gap predictability

For consecutive HL updates $a \to b$ of one instrument, Δ_{HL} is the HL log-mid change. For shift s:

$$c(s) = \text{corr}\Big(\Delta_{\text{HL}},\ \log m^{\text{BN}}(t_b + s) - \log m^{\text{BN}}(t_a + s)\Big). \tag{42}$$

A peak at $s < 0$ means Binance moves first. The peak location *includes* any HL-to-Binance clock offset. **Within-gap:** regress Δ_{HL} on the Binance return over $(t_a,\ t_b - \delta]$, known δ ms before block b, without intercept on train. Then $R^2_{\text{OOS}} = 1 - \sum(\Delta_{\text{HL}} - \hat{b}\, r)^2 / \sum \Delta^2_{\text{HL}}$ is measured against a zero forecast.

16.4 Event study

A Binance *jump* at τ is an update whose 50 ms log-mid return is in the top 0.5% for that pair, keeping the first in any 2 s stretch. For each jump with sign s_τ, both mids are sampled on $\tau + u$, $u \in [-1000, 2000]$ ms:

$$\Pi(u) = \frac{1}{J} \sum_{\tau} s_\tau \cdot 10^4 \big[\log m(\tau + u) - \log m(\tau - 1000)\big] \quad \text{(signed bps)}, \tag{43}$$

with $t_{50} = \min\{u \geq -200 : \Pi(u) \geq \tfrac{1}{2}\Pi(2000)\}$, and $\text{lag} = t^{\text{HL}}_{50} - t^{\text{BN}}_{50}$. HL can only move at its own blocks, so its Π is a staircase averaged over events. The jump threshold is a quantile of the same sample, so the event count is fixed by construction.

Part III

Moving the NUFFT analysis onto the real data

This is what the maths above implies for the next step; no analysis is done here.

1. **The comb to mask is HL's block clock, not Binance's 100 ms.** Every HL event sits on a visible block. With $\bar{\beta} \approx 67\,\text{ms}$ the comb is at $f_h \approx h \times 14.9\,\text{Hz}$. If the clock were a rigid lattice with Gaussian jitter $\sigma = 1.5\,\text{ms}$, (21) damps harmonic h by $e^{-2\pi^2\sigma^2 f_h^2}$: 0.99, 0.96, 0.78, 0.37 for $h = 1, 2, 5, 10$. But 35% of blocks are off the core (M2b), so the lattice phase wanders and the comb will be *broadened*. How much depends on how phase-locked the clock is ($\rho(1) = -0.2$ suggests partly). Periods shorter than about two blocks ($\sim 135\,\text{ms}$) on HL are unobservable.

2. **Binance timestamps.** Use trade time T, not event time E (which is about 150 ms later and batched). bookTicker is event-driven, so it has no 100 ms grid; but check `sampling_report` anyway.

3. **Event definition.** Binance `aggTrade` already merges fills of one taker order; HL prints every fill. Group HL fills by `hash` (one taker order) before counting, or a single sweep becomes a burst that inflates $\kappa(f)$ in (18).

4. **Gaps in the record (masking).** With an observation window $w(t) \in \{0, 1\}$ (0 during reconnects), $\mathbb{E}X(f) = \int \lambda(t) w(t) e^{-2\pi \mathrm{i} f t}\,\mathrm{d}t$. The window's own transform $\hat{w}$ is *convolved* into the spectrum, so a few short gaps add small sidelobes. Normalise by N in the kept window, and drop replayed trades.

5. **Resolution and cycles.** 7.2 h gives a resolution of $1/T = 3.9 \times 10^{-5}\,\text{Hz}$. Requiring ≥ 10 cycles (`rayleigh_test`) limits periods to $\leq 43\,\text{min}$. For a 1 s band to 5 min, $M \approx T(1 - 1/300) \approx 2.6 \times 10^4$, so $z^* \approx 14.8$ at 1%. That's before the burst inflation κ, which is why the block-permutation null matters.

6. **Frequency grid.** Use step $\leq 1/(4T) \approx 10^{-5}\,\text{Hz}$ around named candidates (or the FFT/zero-padded grid), and keep band edges off comb harmonics (§8).

7. **Phase bias from reaction delay.** A burst of reactions after a scheduled order shifts the measured phase by $\arg \gamma(f)$ in (19). Phase is best read from the first fill of each taker order.

8. **More days, weekdays.** One Sunday morning holds few institutional schedules. Stability across days, the same phase every day, is the strongest test in §7, and it needs several days.